

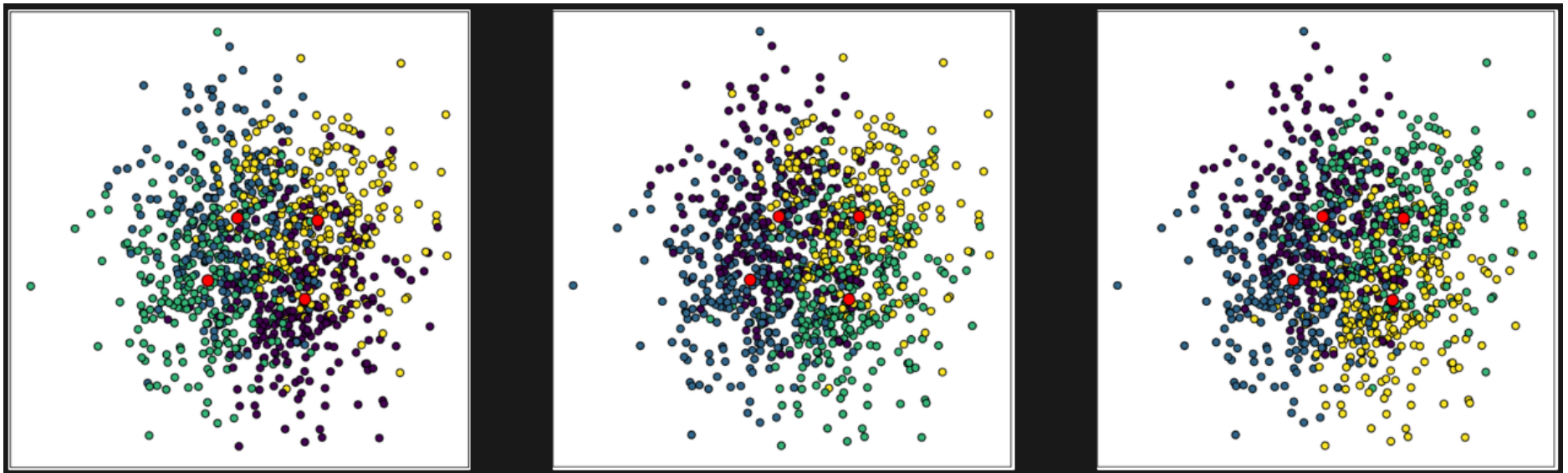
Tsomakian Konstantinos Exercise 2

In this report clusters are computed using 3 different methods, k means clustering, clustering using the mixture of gaussians with random initialization and clustering using the mixture of gaussians initialized by the centroids and means computed from the kmeans algorithm.

The folloing dataset was used:

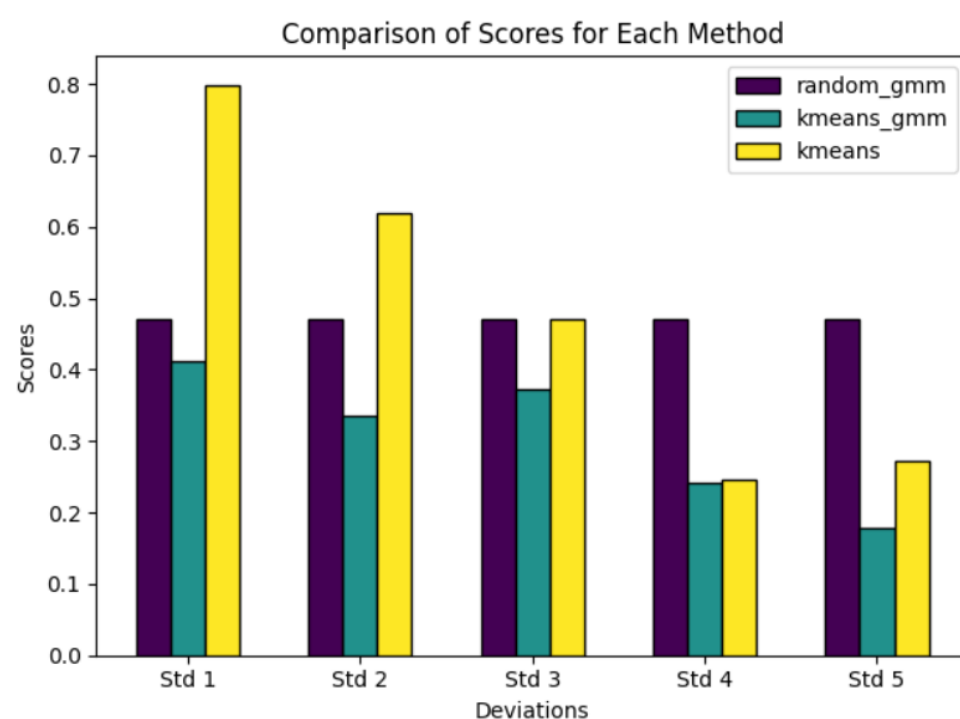
```
from sklearn.datasets import make_blobs
X, y = make_blobs(n_samples=1000, n_features=10, centers=4, cluster_std=3.0, random_state=78)
```

All three algorithms result in similar looking clusters:



From left to right:gmm with random initialization, gmm with kmeans initialization, kmeans

To test how the algorithms perform I computed the silhouette metric for each clustering for standard deviations of the make_blobs function ranging from 1 to 5. The results are shown in the following barplot



What I can tell from the barplot is that although kmeans seem to perform better in the dataset with low deviation gmm algorithm performs better in the dataset with higher standard deviation.

Problems I had(and still have):

1) GMM algorithm seems to not be consistent and I can't tell the reason(The plot on this report is kind of the desired one after some runs)

2) GMM algorithm in random runs raises errors

3) If i got it right from sources I reached out to, when computing the GMM log likelihood the comparison between the log likelihoods is often set to 3 to 5 decimal. When I tried this it seemed that the iterations where not enough to compute clusters.